

EXPERIMENT 17: PERFORM EDGE DETECTION USING SOBEL MATRIX ALONG X AXIS

Aim

To detect image edges along the X-axis using the Sobel operator in Python and OpenCV.

Algorithm

1. Import OpenCV.
2. Read the input image.
3. Convert the image to grayscale.
4. Apply Sobel operator along the X-axis.
5. Convert the result to an 8-bit image.
6. Display and save the output image.

Python Code

```
import cv2

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)

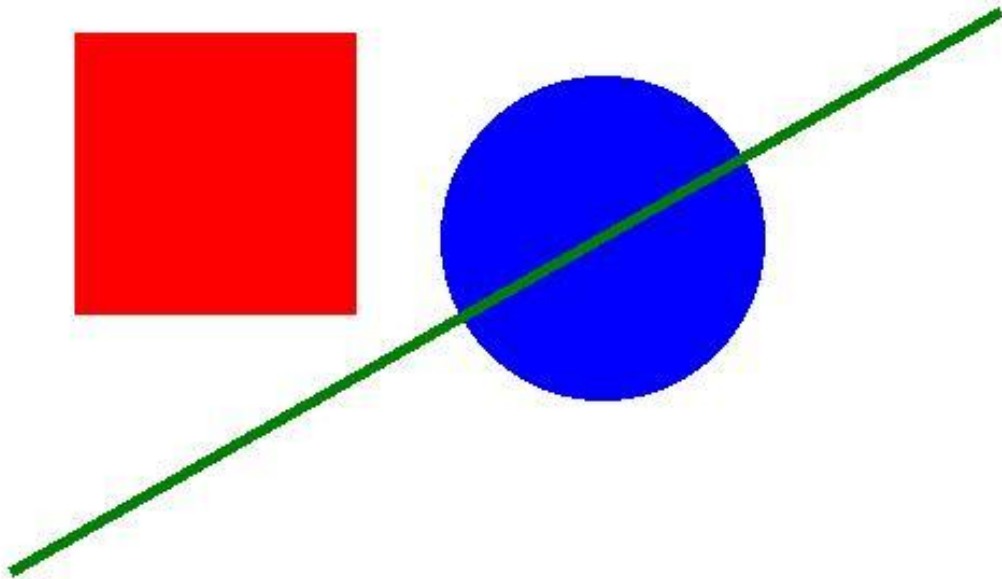
cv2.imwrite("sobel_x_output.jpg", sobel_x)

cv2.imshow("Original Image", img)
cv2.imshow("Sobel X", sobel_x)

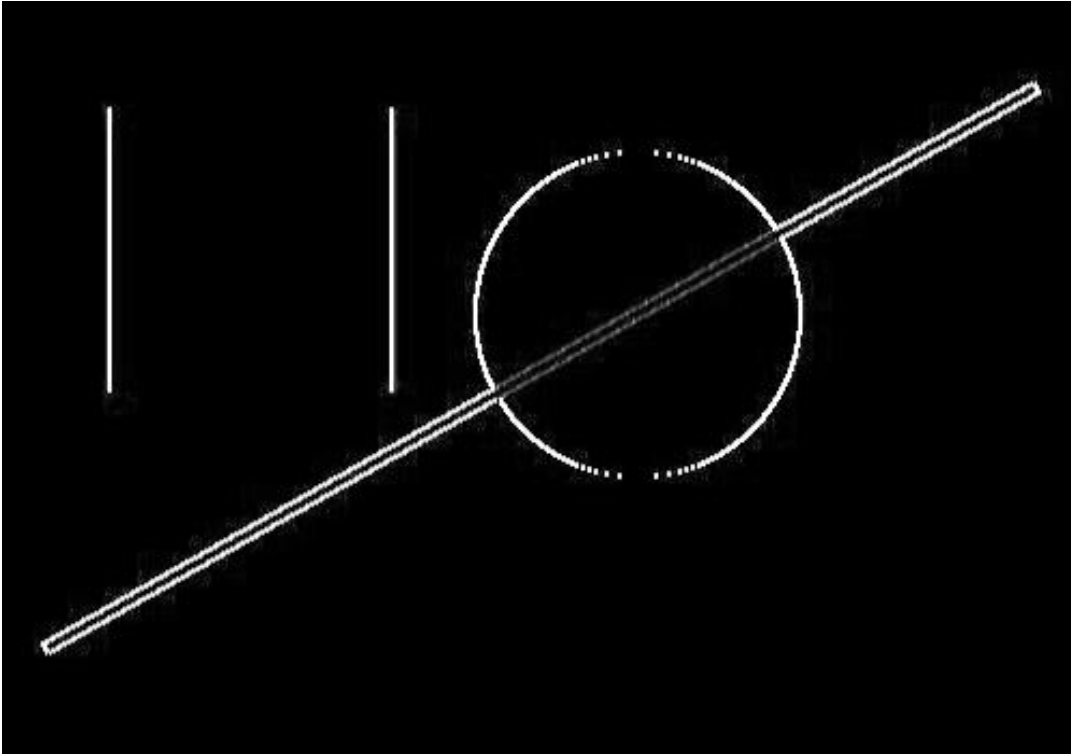
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the edges along the X-axis were successfully detected using the Sobel operator.